

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $\lim_{x \rightarrow a} \frac{2x - \sqrt{x^2 + 3a^2}}{\sqrt{x+a} - \sqrt{2a}} = \sqrt{2}$ (where $a \in \mathbb{R}^+$), then a is equal to -

(A) $\frac{1}{3}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{9}$

MLT217

2. $\lim_{x \rightarrow 0} \frac{\ell n(\sin 3x)}{\ell n(\sin x)}$ is equal to

(A) 0 (B) 1 (C) 2 (D) Non existent

MLT218

3. $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) D.N.E.

MLT219

4. $\lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^3} - \sqrt{3+x^2}}{x-1}$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{6}$

MLT220

5. $\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}}$ is equal to

(A) 1 (B) 100 (C) 200 (D) 10

MLT221

6. $\lim_{x \rightarrow \infty} (\sqrt{x^2 - 2x - 1} - \sqrt{x^2 - 7x + 3})$ is equal to

(A) $-\frac{5}{2}$ (B) $\frac{5}{2}$ (C) 0 (D) D.N.E

MLT222

7. If $\lim_{n \rightarrow \infty} (\sqrt{2n^2 + n} - \lambda \sqrt{2n^2 - n}) = \frac{1}{\sqrt{2}}$ (where λ is a real number), then-

(A) $\lambda = 1$ (B) $\lambda = -1$ (C) $\lambda = \pm 1$ (D) $\lambda \in (-\infty, 1)$

MLT223

8. Let $U_n = \frac{n!}{(n+2)!}$ where $n \in \mathbb{N}$. If $S_n = \sum_{r=1}^n U_r$ then $\lim_{n \rightarrow \infty} S_n$ equals

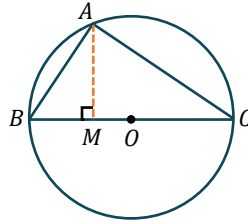
(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) Non existent

MLT224

9. For $n \in \mathbb{N}$, let $a_n = \sum_{k=1}^n 2k$ and $b_n = \sum_{k=1}^n (2k-1)$. Then $\lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{b_n})$ is equal to-
- (A) 1 (B) $\frac{1}{2}$ (C) 0 (D) 2 MLT225
10. Let $P_n = \prod_{k=2}^n \left(1 - \frac{1}{k+1} C_2\right)$. If $\lim_{n \rightarrow \infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a+b)$ is
- (A) 4 (B) 8 (C) 10 (D) 12 MLT226
11. $\lim_{x \rightarrow -1} \frac{\cos 2 - \cos 2x}{x^2 - |x|}$ is equal to
- (A) 0 (B) $\cos 2$ (C) $2\sin 2$ (D) $\sin 1$ MLT227
12. $\lim_{x \rightarrow 0} \left(\left[\frac{-5\sin x}{x} \right] + \left[\frac{6\sin x}{x} \right] \right)$ (where $[.]$ denotes greatest integer function) is equal to -
- (A) 0 (B) -12 (C) 1 (D) 2 MLT228
13. Let $f(x) = \left[\frac{\sin x}{x} \right] + \left[\frac{2\sin 2x}{x} \right] + \dots + \left[\frac{10\sin 10x}{x} \right]$ (where $[y]$ is the largest integer $\leq y$). The value of $\lim_{x \rightarrow 0} f(x)$ equals
- (A) 55 (B) 164 (C) 165 (D) 375 MLT229
14. Let $f(x) = \frac{\sin\{x\}}{x^2 + ax + b}$. If $f(5^+)$ & $f(3^+)$ exists finitely and are not zero, then the value of $(a+b)$ is (where $\{.\}$ represents fractional part function)-
- (A) 7 (B) 10 (C) 11 (D) 20 MLT230
15. $\lim_{x \rightarrow 0} \frac{|\cos(\sin(3x))| - 1}{x^2}$ equals
- (A) $-\frac{9}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{3}{2}$ (D) $\frac{9}{2}$ MLT231
16. Let $a = \min\{x^2 + 2x + 3, x \in \mathbb{R}\}$ and $b = \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$. Then value of $\sum_{r=0}^n a^r \cdot b^{n-r}$ is :
- (A) $\frac{2^{n+1} - 1}{3 \cdot 2^n}$ (B) $\frac{2^{n+1} + 1}{3 \cdot 2^n}$ (C) $\frac{4^{n+1} - 1}{3 \cdot 2^n}$ (D) $\frac{4^{n+1} + 1}{3 \cdot 2^n}$ MLT232

Limits

17. Let BC is diameter of a circle centred at O . Point A is a variable point, moving on the circumference of circle. If $BC = 1$ unit, then $\lim_{A \rightarrow B} \frac{BM}{(\text{Area of sector } OAB)^2}$ is equal to -



- (A) 1 (B) 2 (C) 4 (D) 16 MLT233

18. $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 2} \right)^x$ is equal to

- (A) 1 (B) e (C) $\frac{1}{e^2}$ (D) e^2

MLT234

19. $\lim_{x \rightarrow 0} (1 + \sin x)^{\cos x}$ is equal to

- (A) 0 (B) e (C) 1 (D) $\frac{1}{e}$

MLT235

20. $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}}$ is equal to

- (A) e^a (B) e^{ab} (C) e^b (D) $e^{a/b}$

MLT236

21. $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} + x \right) \right)^{1/x}$ is equal to

- (A) e^{-2} (B) $\frac{1}{e}$ (C) e (D) e^2

MLT237

22. $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to

- (A) 5 (B) 4 (C) 0 (D) D.N.E.

MLT238

23. $\lim_{x \rightarrow \infty} \left(\frac{1^{1/x} + 2^{1/x} + 3^{1/x} + \dots + n^{1/x}}{n} \right)^{nx}$, $n \in \mathbb{N}$ is equal to

- (A) $n!$ (B) 1 (C) $\frac{1}{n!}$ (D) 0

MLT239

24. If $\lim_{x \rightarrow \lambda} \left(2 - \frac{\lambda}{x} \right)^{\lambda \tan \left(\frac{\pi x}{2\lambda} \right)} = \frac{1}{e}$, then λ is equal to -

- (A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) $-\frac{2}{\pi}$

MLT240

25. If $\lim_{x \rightarrow 0} (1 + ax + bx^2)^{2/x} = e^e$, then

(A) $a = \frac{3}{2}$ and $b \in \mathbb{R}$

(B) $a = \frac{3}{2}$ and $b \in \mathbb{R}^+$

(C) $a = 0$ and $b = 1$

(D) $a = 1$ and $b = 0$

MLT241

26. If $f(x)$ is a polynomial of least degree, such that $\lim_{x \rightarrow 0} \left(1 + \frac{f(x) + x^2}{x^2} \right)^{1/x} = e^2$, then $f(2)$ is -

(A) 2

(B) 8

(C) 10

(D) 12

MLT242

27. Let $f(x) = \frac{\tan x}{x}$, then the value of $\lim_{x \rightarrow 0} \left([f(x)] + x^2 \right)^{\frac{1}{\{f(x)\}}}$ is equal to (where $[.]$, $\{.\}$ denotes greatest integer function and fractional part functions respectively) -

(A) e^{-3}

(B) e^3

(C) e^2

(D) non-existent

MLT243

28. $\lim_{n \rightarrow \infty} \frac{e^n}{\left(1 + \frac{1}{n}\right)^{n^2}}$ equals -

(A) 1

(B) $\frac{1}{2}$

(C) e

(D) $\sqrt{e}$

MLT244

29. If $f(x)$ is odd linear polynomial with $f(1) = 1$, then $\lim_{x \rightarrow 0} \frac{2^{f(\tan x)} - 2^{f(\sin x)}}{x^2 f(\sin x)}$ is

(A) 1

(B) $\ln 2$

(C) $\frac{1}{2} \ln 2$

(D) $\cos 2$

MLT245

30. $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = 1$ then

(A) $a = \frac{-5}{2}$

(B) $a = \frac{-3}{2}, b = \frac{-1}{2}$

(C) $a = \frac{-3}{2}, b = \frac{-5}{2}$

(D) $a = \frac{-5}{2}, b = \frac{-3}{2}$

MLT246

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let $f(x) = \begin{cases} \sin x; & \text{where } x = \text{integer} \\ 0; & \text{otherwise} \end{cases}$: $g(x) = \begin{cases} x^2 + 1; & x \neq 0, 2 \\ 4; & x = 0 \\ 5; & x = 2 \end{cases}$, then
- (A) $\lim_{x \rightarrow 0} g(f(x)) = 4$ (B) $\lim_{x \rightarrow 0} f(g(x)) = 0$
 (C) $\lim_{x \rightarrow 1} f(g(x)) = 0$ (D) $\lim_{x \rightarrow 1} g(f(x)) = 5$

MLT055

2. Let $f(x) = \begin{cases} 1 + \frac{2x}{a}, & 0 \leq x < 1 \\ ax, & 1 \leq x < 2 \end{cases}$, if $\lim_{x \rightarrow 1} f(x)$ exists, then value(s) of a is :
- (A) 1 (B) -1 (C) 2 (D) -2

MLT056

3. If $A = \lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B = \lim_{x \rightarrow 0} \frac{[x]}{x}$, then (where $[.]$ denotes greatest integer function)-
- (A) $A = 1$ (B) A does not exist
 (C) $B = 0$ (D) $B = 1$

MLT058

4. The value(s) of 'n' for which $\lim_{x \rightarrow 1} \frac{e^{x-1} - x}{(x-1)^n}$ exists is/are -
- (A) 1 (B) 2 (C) 3 (D) 4

MLT060

5. $\lim_{x \rightarrow c} f(x)$ does not exist when $\text{LHL} \neq \text{RHL}$ (where $[x]$ is the step-up function, $\{x\}$ is the fractional part function of x & $\text{sgn}(x)$ denotes signum function), then-
- (A) $f(x) = [x] - [2x - 1]; c = 3$ (B) $f(x) = [x] - x, c = 1$
 (C) $f(x) = \{x\}^2 - \{-x\}^2, c = 0$ (D) $f(x) = \frac{\tan(\text{sgn } x)}{\text{sgn } x}, c = 0$

MLT062

6. If $f(x) = \frac{\cos 2 - \cos 2x}{x^2 - |x|}$, then
- (A) $\lim_{x \rightarrow -1} f(x) = 2\sin 2$ (B) $\lim_{x \rightarrow 1} f(x) = 2\sin 2$
 (C) $\lim_{x \rightarrow -1} f(x) = 2\cos 2$ (D) $\lim_{x \rightarrow 1} f(x) = 2\cos 2$

MLT063

7. Which of the following limits equal to $\frac{1}{2}$
- (A) $\lim_{n \rightarrow \infty} \left(\frac{1}{1.3} + \frac{1}{3.5} + \dots + \frac{1}{(2n-1)(2n+1)} \right)$
 (C) $\lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n)$

(B) $\lim_{x \rightarrow \infty} \left[\frac{3x^2}{2x+1} - \frac{(2x-1)(3x^2+x+2)}{4x^2} \right]$
 (D) $\lim_{n \rightarrow \infty} \frac{(n+2)! + (n+1)!}{(n+2)! - (n+1)!}$

MLT247

8. Which of the following limit tends to unity ?

(A) $\lim_{x \rightarrow 0} \frac{1 - \cos x + 2 \sin x - \sin^3 x - x^2 + 3x^4}{\tan^3 x - 6 \sin^2 x + x - 5x^3}$

(B) $\lim_{x \rightarrow \infty} \frac{x}{[x]}$

(C) $\lim_{x \rightarrow \infty} \frac{1}{(\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x})}$

(D) $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x}}{\sqrt{x + \sqrt{x + \sqrt{x}}}} \right)$

MLT248

9. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{(\tan x)^{2n} + x^2}{\sin^2 x + (\tan x)^{2n}} \right) & ; x \neq 0 \\ 1 & ; x = 0 \end{cases}$, $n \in \mathbb{N}$. Which of the following holds

good ?

(A) $f\left(-\frac{\pi}{4}\right) = f\left(\frac{\pi}{4}\right)$

(B) $f\left(-\frac{\pi}{4}\right) = f\left(-\frac{\pi^+}{4}\right)$

(C) $f\left(\frac{\pi^-}{4}\right) = f\left(\frac{\pi^+}{4}\right)$

(D) $f(0^+) = f(0) = f(0^-)$

MLT249

10. Let $f(x) = \begin{cases} \frac{\tan^2 \{x\}}{x^2 - [x]^2} & \text{for } x > 0 \\ 1 & \text{for } x = 0, \text{ where } [x] \text{ is the step up function and } \{x\} \text{ is the fractional part} \\ \sqrt{\{x\} \cot \{x\}} & \text{for } x < 0 \end{cases}$

function of x , then-

(A) $\lim_{x \rightarrow 0^+} f(x) = 1$

(B) $\lim_{x \rightarrow 0^-} f(x) = 1$

(C) $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = 1$

(D) $\lim_{x \rightarrow 0^+} f(x) = 2$

MLT250

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 11 to 13

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$,

Where $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$, then

11. $f(x)$ is

(A) e^{-x^2}

(B) $e^{\frac{-x^2}{2}}$

(C) e^{x^2}

(D) $e^{\frac{x^2}{2}}$

MLT064

12. $g(x)$ is

(A) $-x^2$

(B) x^2

(C) x^4

(D) $-x^4$

MLT065

13. Number of solutions of equation $f(x) + g(x) = 0$ is

(A) 2

(B) 4

(C) 0

(D) 1

MLT066

Paragraph for Question No. 14 to 16

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$, where a, b, c are real numbers.

14. If $\lim_{x \rightarrow 0^+} f(x)$ is finite, then the value of $a - b + c$ is

MLT067

15. If $\lim_{x \rightarrow 0^+} f(x) = \ell$ (finite), then the value of ℓ is

MLT068

16. Using the values of a, b, c as found above, the value of $\lim_{x \rightarrow 0^+} x f(x)$ is

MLT069

MATCHING LIST TYPE QUESTION

17. $\lim_{x \rightarrow 0} f(x)$, where $f(x)$ is as in LIST-I, is

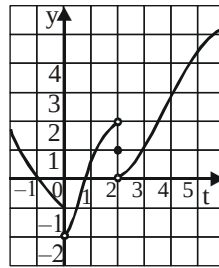
	List-I		List-II
(P)	$f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$ where $[.]$ represents greatest integer function.	(1)	$\frac{\sqrt{2}}{8}$
(Q)	$f(x) = \left[(\min(t^2 + 4t + 6)) \frac{\sin x}{x} \right]$ where $[.]$ represents greatest integer function.	(2)	15
(R)	$f(x) = \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$	(3)	1
(S)	$f(x) = \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$	(4)	$\frac{1}{2}$

- (A) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (1), (S) $\rightarrow$ (4)
 (D) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (3), (S) $\rightarrow$ (1)

MLT070

MATRIX MATCH TYPE QUESTION

18. For the function $g(t)$ whose graph is given, match the entries of column-I to column-II



Column-I

- (P) $\lim_{t \rightarrow 0^+} g(t) + \lim_{t \rightarrow 2^-} g(t)$
 (Q) $\lim_{t \rightarrow 0} g(t) + g(2)$
 (R) $\lim_{t \rightarrow 0} g(t)$
 (S) $\lim_{t \rightarrow 2} g(t)$

Column-II

- (1) $\lim_{t \rightarrow 2^+} g(t)$ **MLT071**
 (2) does not exist **MLT072**
 (3) 0 **MLT073**
 (4) $\lim_{t \rightarrow 4} g(t)$ **MLT074**

19.

Column-I

- (P) $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{4n}\right) \cos\left(\frac{\pi}{4n}\right)$ is equal to
 (Q) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ is equal to
 (R) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$ is equal to
 (S) $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}$ is equal to

Column-II

- (1) 0 **MLT075**
 (2) $\frac{1}{2}$ **MLT076**
 (3) $\frac{\pi}{4}$ **MLT077**
 (4) $\frac{\pi}{180}$ **MLT078**

20.

Column-I

- (P) $\lim_{x \rightarrow \infty} \frac{2^x}{2^x + 1}$ can be equal to
 (Q) $\lim_{x \rightarrow 2} \frac{2 \sin(e^{x-2} - 1)}{\log(x - 1)}$ is equal to
 (R) $\lim_{x \rightarrow e} \frac{(\ln x - 1)e}{x - e}$ is equal to
 (S) $\lim_{x \rightarrow 0} \frac{x(5^x - 1)}{(1 - \cos x)4 \ln 5}$ is equal to

Column-II

- (1) 2 **MLT079**
 (2) 1 **MLT080**
 (3) $\frac{1}{2}$ **MLT081**
 (4) 0 **MLT082**

EXERCISE (S)

1. $\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}}$ is equal to MLT021
2. Let $P_n = \prod_{k=2}^n \left(1 - \frac{1}{k+1} C_2\right)$. If $\lim_{n \rightarrow \infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a + b)$ is MLT022
3. $\lim_{x \rightarrow 0} \left(\left\lfloor \frac{-5 \sin x}{x} \right\rfloor + \left\lceil \frac{6 \sin x}{x} \right\rceil \right)$ (where $[.]$ denotes greatest integer function) is equal to - MLT023
4. Let $f(x) = \frac{\sin\{x\}}{x^2 + ax + b}$. If $f(5^+)$ & $f(3^+)$ exists finitely and are not zero, then the value of $(a + b)$ is (where $\{.\}$ represents fractional part function) - MLT024
5. $\lim_{n \rightarrow \infty} (4^n + 5^n)^{\frac{1}{n}}$ is equal to MLT025
6. $\lim_{x \rightarrow 1} \frac{x^2 - x \ln x + \ln x - 1}{x - 1}$ MLT026
7. $\lim_{x \rightarrow 0} [\ell n(1 + \sin^2 x) \cdot \cot(\ell n^2(1 + x))]$ MLT027
8. If $\lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, then the value of $(a + b + c)$ can be expressed in the lowest form as $\frac{p}{q}$. Find the value of $(p + q)$. MLT028
9. $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1} = \ln k$, where k is equal to MLT029
10. $L = \lim_{x \rightarrow \infty} x^2 \left(\sqrt{\frac{x+2}{x}} - \sqrt[3]{\frac{x+3}{x}} \right)$ then $2L$ equal to MLT030

EXERCISE - JEE (Main) PYQ

1. For each $t \in R$, let $[t]$ be the greatest integer less than or equal to t . Then

$$\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$$

[JEE (Main) 2018]

(1) is equal to 15.

(2) is equal to 120.

(3) does not exist (in R).

(4) is equal to 0.

MLT038

2.
$$\lim_{y \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + y^4}} - \sqrt{2}}{y^4}$$

[JEE (Main) 2019]

(1) exists and equals $\frac{1}{4\sqrt{2}}$

(2) does not exist

(3) exists and equals $\frac{1}{2\sqrt{2}}$

(4) exists and equals $\frac{1}{2\sqrt{2}(\sqrt{2}+1)}$

MLT039

3. For each $x \in R$, let $[x]$ be the greatest integer less than or equal to x . Then $\lim_{x \rightarrow 0^+} \frac{x([x] + |x|)\sin[x]}{|x|}$ is

equal to

(1) $-\sin 1$

(2) 0

(3) 1

(4) $\sin 1$

[JEE (Main) 2019]

MLT040

4. For each $t \in R$, let $[t]$ be the greatest integer less than or equal to t .

Then,
$$\lim_{x \rightarrow 1^+} \frac{(1 - |x| + \sin |1 - x|) \sin \left(\frac{\pi}{2} [1 - x] \right)}{|1 - x| [1 - x]}$$

[JEE (Main) 2019]

(1) equals -1

(2) equals 1

(3) does not exist

(4) equals 0

MLT041

5. Let $[x]$ denote the greatest integer less than or equal to x .

Then
$$\lim_{x \rightarrow 0} \frac{\tan(\pi \sin^2 x) + (|x| - \sin(x[x]))^2}{x^2}$$

[JEE (Main) 2019]

(1) equals π

(2) equals 0

(3) equals $\pi + 1$

(4) does not exist

MLT042

6.
$$\lim_{x \rightarrow \pi/4} \frac{\cot^3 x - \tan x}{\cos(x + \pi/4)}$$
 is :

[JEE (Main) 2019]

(1) 4

(2) $8\sqrt{2}$

(3) 8

(4) $4\sqrt{2}$

MLT043

7.
$$\lim_{x \rightarrow 1^-} \frac{\sqrt{\pi} - \sqrt{2 \sin^{-1} x}}{\sqrt{1 - x}}$$
 equal to:

[JEE (Main) 2019]

(1) $\frac{1}{\sqrt{2\pi}}$

(2) $\sqrt{\frac{\pi}{2}}$

(3) $\sqrt{\frac{2}{\pi}}$

(4) $\sqrt{\pi}$

MLT044

Limits

8. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$ equals: [JEE (Main) 2019]
 (1) $2\sqrt{2}$ (2) $4\sqrt{2}$ (3) $\sqrt{2}$ (4) 4
 MLT045
9. If $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$, then k is : [JEE (Main) 2019]
 (1) $\frac{3}{8}$ (2) $\frac{3}{2}$ (3) $\frac{4}{3}$ (4) $\frac{8}{3}$
 MLT046
10. If $\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$, then $a + b$ is equal to :- [JEE (Main) 2019]
 (1) -7 (2) -4 (3) 5 (4) 1
 MLT047
11. $\lim_{x \rightarrow 2} \frac{3^x + 3^{3-x} - 12}{3^{-x/2} - 3^{1-x}}$ is equal to. [JEE (Main) 2020]
 MLT048
12. $\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ is equal to [JEE (Main) 2020]
 (1) $\frac{1}{e}$ (2) e^2 (3) e (4) $\frac{1}{e^2}$
 MLT049
13. If $\lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - n}{x - 1} = 820, (n \in \mathbb{N})$ then the value of n is equal to. [JEE (Main) 2020]
 MLT050
14. $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} + x \right) \right)^{1/x}$ is equal to : [JEE (Main) 2020]
 (1) 2 (2) e (3) 1 (4) e^2
 MLT051
15. $\lim_{n \rightarrow \infty} \tan \left\{ \sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right\}$ is equal to. [JEE (Main) 2021]
 MLT052
16. The value of $\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(x - [x]^2) \cdot \sin^{-1}(x - [x]^2)}{x - x^3}$, where $[x]$ denotes the greatest integer $\leq x$ is : [JEE (Main) 2021]
 (1) π (2) 0 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{2}$
 MLT097
17. If $\lim_{x \rightarrow \infty} (\sqrt{x^2 - x + 1} - ax) = b$, then the ordered pair (a, b) is: [JEE (Main) 2021]
 (1) $\left(1, \frac{1}{2}\right)$ (2) $\left(1, -\frac{1}{2}\right)$ (3) $\left(-1, \frac{1}{2}\right)$ (4) $\left(-1, -\frac{1}{2}\right)$
 MLT098

18. $\lim_{x \rightarrow 0} \frac{\sin^2(\pi \cos^4 x)}{x^4}$ is equal to : [JEE (Main) 2021]
 (1) π^2 (2) $2\pi^2$ (3) $4\pi^2$ (4) 4π MLT099

19. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\tan^2 x \left((2\sin^2 x + 3\sin x + 4)^{\frac{1}{2}} - (\sin^2 x + 6\sin x + 2)^{\frac{1}{2}} \right) \right)$ is equal to [JEE (Main) 2022]
 (1) $\frac{1}{12}$ (2) $-\frac{1}{18}$ (3) $-\frac{1}{12}$ (4) $-\frac{1}{6}$ MLT100

20. $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4}$ is equal to : [JEE (Main) 2022]
 (1) $\frac{1}{3}$ (2) $\frac{1}{4}$ (3) $\frac{1}{6}$ (4) $\frac{1}{12}$ MLT101

21. $\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{\sin(\cos^{-1} x) - x}{1 - \tan(\cos^{-1} x)}$ is equal to : [JEE (Main) 2022]
 (1) $\sqrt{2}$ (2) $-\sqrt{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) $-\frac{1}{\sqrt{2}}$ MLT102

22. $\lim_{x \rightarrow \infty} \frac{(\sqrt{3x+1} + \sqrt{3x-1})^6 + (\sqrt{3x+1} - \sqrt{3x-1})^6}{(x + \sqrt{x^2-1})^6 + (x - \sqrt{x^2-1})^6} x^3$ [JEE (Main) 2023]
 (1) is equal to 9 (2) is equal to 27 (3) does not exist (4) is equal to $\frac{27}{2}$ MLT103

23. The value of $\lim_{n \rightarrow \infty} \frac{1+2-3+4+5-6+\dots+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ is : [JEE (Main) 2023]
 (1) $\frac{\sqrt{2}+1}{2}$ (2) $3(\sqrt{2}+1)$ (3) $\frac{3}{2}(\sqrt{2}+1)$ (4) $\frac{3}{2\sqrt{2}}$ MLT104

24. The set of all values of a for which $\lim_{x \rightarrow a} ([x-5] - [2x+2]) = 0$, where $[x]$ denotes the greater integer less than or equal to x is equal to [JEE (Main) 2023]
 (1) $(-7.5, -6.5)$ (2) $[-7.5, -6.5]$ (3) $[-7.5, -6.5]$ (4) $[-7.5, -6.5]$ MLT105

25. $\lim_{t \rightarrow 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \dots + n^{\frac{1}{\sin^2 t}} \right)^{\sin^2 t}$ is equal to [JEE (Main) 2023]
 (1) $n^2 + n$ (2) n (3) $\frac{n(n+1)}{2}$ (4) n^2 MLT106

26. If $\lim_{x \rightarrow 0} \frac{e^{ax} - \cos(bx) - \frac{cxe^{-cx}}{2}}{1 - \cos(2x)} = 17$, then $5a^2 + b^2$ is equal to

(1) 72

(2) 76

(3) 68

(4) 64

[JEE (Main) 2023]

MLT107

27. If $\lim_{x \rightarrow 1} \frac{(5x+1)^{1/3} - (x+5)^{1/3}}{(2x+3)^{1/2} - (x+4)^{1/2}} = \frac{m\sqrt{5}}{n(2n)^{2/3}}$, where $\gcd(m, n) = 1$, then $8m + 12n$ is equal to ____

[JEE (Main) 2024]

MLT146

28. Let $a > 0$ be a root of the equation $2x^2 + x - 2 = 0$. If $\lim_{x \rightarrow \frac{1}{a}} \frac{16(1 - \cos(2 + x - 2x^2))}{(1 - ax^2)} = \alpha + \beta\sqrt{17}$, where $\alpha, \beta \in \mathbb{Z}$ then $\alpha + \beta$ is equal to ____.

[JEE (Main) 2024]

MLT147

29. $\lim_{n \rightarrow \infty} \frac{(1^2 - 1)(n-1) + (2^2 - 2)(n-2) + \dots + ((n-1)^2 - (n-1)) \cdot 1}{(1^3 + 2^3 + \dots + n^3) - (1^2 + 2^2 + \dots + n^2)}$ is equal to:

[JEE (Main) 2024]

(1) $\frac{2}{3}$

(2) $\frac{1}{3}$

(3) $\frac{3}{4}$

(4) $\frac{1}{2}$

MLT148

30. If $\alpha = \lim_{x \rightarrow 0^+} \left(\frac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}} \right)$ and $\beta = \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{2} \cot x}$ are the roots of the quadratic equation $ax^2 + bx - \sqrt{e} = 0$, then $12 \log_e(a + b)$ is equal to ____.

[JEE (Main) 2024]

MLT149

EXERCISE - JEE (Advanced) PYQ

1. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then - [IIT JEE 2009]
 (A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$
MLT083
2. If $\lim_{x \rightarrow 0} \left[1 + x \ln(1 + b^2) \right]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is- [IIT JEE 2011]
 (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$
MLT084
3. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then - [IIT JEE 2012]
 (A) $a = 1, b = 4$ (B) $a = 1, b = -4$
 (C) $a = 2, b = -3$ (D) $a = 2, b = 3$
MLT085
4. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[5]{1+a} - 1) = 0$ where $a > -1$. Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are [IIT JEE 2012]
 (A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1 (C) $-\frac{7}{2}$ and 2 (D) $-\frac{9}{2}$ and 3
MLT086
5. The largest value of the non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is [JEE (Advanced) 2014]
MLT087
6. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2} \right)$ then the value of $\frac{m}{n}$ is [JEE (Advanced) 2015]
MLT088
7. Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals [JEE (Advanced) 2016]
MLT089
8. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then [JEE (Advanced) 2017]
 (C) $\lim_{x \rightarrow 1^-} f(x) = 0$ (D) $\lim_{x \rightarrow 1^+} f(x) = 0$
 (A) $\lim_{x \rightarrow 1^+} f(x)$ does not exist (B) $\lim_{x \rightarrow 1^-} f(x)$ does not exist
MLT090

9. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as

$$f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1+(x+j)(x+j-1)} \right) \text{ for all } x \in (0, \infty).$$

(Here, the inverse trigonometric function $\tan^{-1} x$ assume values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$)

Then, which of the following statement(s) is (are) TRUE ?

[JEE (Advanced) 2018]

(A) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$

(B) $\sum_{j=1}^{10} (1 + f_j'(0)) \sec^2(f_j(0)) = 10$

(C) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$

(D) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

MLT091

10. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a function. We say that f has

PROPERTY 1 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite, and

PROPERTY 2 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h^2}$ exists and is finite.

Then which of the following options is/are correct ?

[JEE (Advanced) 2019]

(A) $f(x) = x|x|$ has PROPERTY 2

(B) $f(x) = x^{2/3}$ has PROPERTY 1

(C) $f(x) = \sin x$ has PROPERTY 2

(D) $f(x) = |x|$ has PROPERTY 1

MLT092

11. Let e denote the base of the natural logarithm. The value of the real number a for which the right-

hand limit $\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - e^{-1}}{x^a}$ is equal to a nonzero real number, is.

[JEE (Advanced) 2020]

MLT093

12. The value of the limit $\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2\sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2} \cos 2x + \cos \frac{3x}{2}\right)}$ is

[JEE (Advanced) 2020]

MLT094

13. Let α be a positive real number. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : (\alpha, \infty) \rightarrow \mathbb{R}$ be the functions defined by

$$f(x) = \sin\left(\frac{\pi x}{12}\right) \text{ and } g(x) = \frac{2 \log_e(\sqrt{x} - \sqrt{\alpha})}{\log_e(e^{\sqrt{x}} - e^{\sqrt{\alpha}})}. \text{ Then the value of } \lim_{x \rightarrow \alpha^+} f(g(x)) \text{ is } \underline{\hspace{2cm}}.$$

[JEE (Advanced) 2022]

MLT095

14. If $\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1 - x^3)^{\frac{1}{3}} + \left((1 - x^2)^{\frac{1}{2}} - 1 \right) \sin x}{x \sin^2 x}$, then the value of 6β is. [JEE (Advanced) 2022]

MLT096

15. Let $k \in \mathbb{R}$. If $\lim_{x \rightarrow 0^+} (\sin(\sin kx) + \cos x + x)^{\frac{2}{x}} = e^6$, then the value of k is [JEE (Advanced) 2024]
- (A) 1 (B) 2 (C) 3 (D) 4

MLT150

16. Let S be the set of all $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ such that

$$\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\log_e x)^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\log_e(1+x))^\beta} = 0.$$

Then which of the following is (are) correct?

[JEE (Advanced) 2024]

- (A) $(-1, 3) \in S$ (B) $(-1, 1) \in S$ (C) $(1, -1) \in S$ (D) $(1, -2) \in S$

MLT151

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	C	B	A	A	C	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	D	A	A	C	D	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	A	C	A	D	B	D	C	D

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	B,C	B,C	A,B	B,C	A,B	A,C	B,D	A,D	A,C
Que.	11	12	13	14	15	16	17	18		
Ans.	B	A	A	-1.00	-0.33	0	B	$P \rightarrow 1,3; Q \rightarrow 1,3; R \rightarrow 2; S \rightarrow 2$		
Que.	19				20					
Ans.	$P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$				$P \rightarrow 2; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$					

EXERCISE (S)

1.	100	2.	4	3.	0	4.	7	5.	5
6.	2	7.	1	8.	167	9.	4	10.	1

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	1	4	4	3	3	2	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	36	4	40	4	1	4	2	3	1	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	3	1	2	3	100	170	2	6

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	D	B	B	0	2	7	A,C	D	B,D
Que.	11	12	13	14	15	16				
Ans.	1.00	8	0.50	5	B	B,C				

Important Notes